

10. The Equator, Tropic of Cancer and Tropic of Capricorn, all the three pass through which one of the following Continents ?

- (a) Africa (b) Asia
(c) North America (d) South America

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (a)

The Equator, Tropic of Cancer and Tropic of Capricorn all the three pass through the African continent. The Equator and the Tropic of Capricorn pass through the South American continent.

11. Intertropical Convergence Zone (ITCZ), a low-pressure zone, is located.

- (a) between trade winds and westerlies belt.
(b) between westerlies and polar winds belt
(c) nearby the poles
(d) at the equator

M.P. P.C.S. (Pre) 2018

Ans. (d)

The Intertropical Convergence Zone is a low-pressure zone located at the equator.

iv. The Tropic of Cancer

*The Tropic of Cancer is **23°30'** degrees latitude north of the equator. Due to the axial tilt of the earth and its revolution, the Tropic of Cancer (**23°30'N** latitude) starts getting oriented towards the sun by 21st March. *Three months later i.e. by **21st June** the Tropic of Cancer is aligned in a way that it receives the vertical rays of the Sun. Hence **21st June** is known as the day of the summer solstice in Northern Hemisphere. *The Tropic of Cancer passes through the following countries/ regions of the world—

- (1) Algeria
- (2) Niger
- (3) Libya
- (4) Egypt
- (5) Saudi Arabia
- (6) United Arab Emirates
- (7) Oman
- (8) India
- (9) Bangladesh
- (10) Myanmar
- (11) China
- (12) Taiwan
- (13) Mexico
- (14) Bahamas
- (15) Mauritania
- (16) Mali
- (17) Western Sahara

22nd December, when the sun is vertically overhead of The Tropic of Capricorn it is called winter solstice. Since the Earth is inclined at an angle of 23°30', when the sun's rays fall at a perpendicular angle (90°) at the Tropic of Capricorn, then at the same time the Tropic of cancer will witness an inclination of 90°–47°= 43°. *The angular difference between the Tropic of Cancer and the Tropic of Capricorn is 23°30' + 23°30' = 47°.

1. At the Winter Solstice the noon altitude of the Sun at the Tropic of Cancer is :

- (a) 23.5° (b) 43.0°
(c) 47.0° (d) 66.5°

U.P.P.C.S. (Pre) 1997

Ans. (b)

The winter solstice falls around 22 December when rays of the Sun fall perpendicular upon the Tropic of Capricorn because the Earth is tilted at its axis by 23°30'. Therefore when at the Tropic of Capricorn the elevation of the Sun is 90°, the elevation of the Sun at the Tropic of Cancer will be 90° – 47° = 43° at the same time. The difference between the Tropic of Cancer and Tropic of Capricorn is 23°30' + 23°30' = 47°.

2. The Tropic of Cancer does not pass through –

- (a) Egypt (b) India
(c) Iran (d) Myanmar

U.P.P.C.S. (Pre) 2005

Ans. (c)

The Tropic of Cancer passes through the following countries/ regions of the world –

Mexico	Saudi Arabia
Bahamas	United Arab Emirates
Mauritania	Oman
Mali	India
Algeria	Bangladesh
Niger	Myanmar
	China
Libya	Taiwan
Egypt	Western Sahara

Hence, it is clear that the Tropic of Cancer does not pass through Iran. Therefore (c) is the correct answer.

v. The Tropic of Capricorn

The 23°30' South latitude in the Southern Hemisphere is known as the Tropic of Capricorn. On 21/22 December the sun shines vertically overhead the Tropic of Capricorn, and it is the longest day in the Southern Hemisphere receiving sunlight for the maximum duration. *During this time there is low air pressure in the Southern Hemisphere and thus the